

FC7xxx SENT User Manual

Rev.0.1

Revision History

Revision	Date	Changes
0.1	2023/12/15	Initial release

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Chapter 1 SENT Introduction

1.1 Requirement Tracing

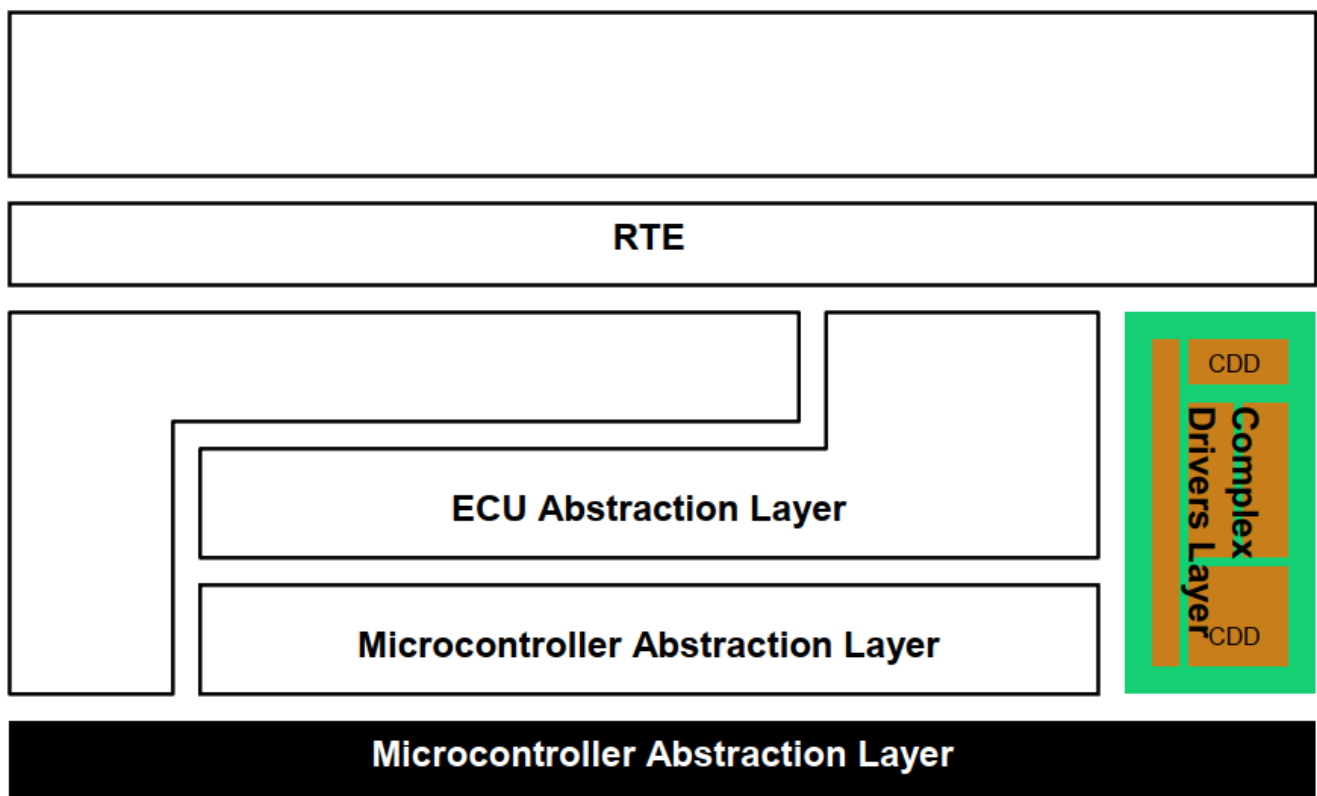
The design of this module follows the specifications of the complex driver specified in AUTOSAR Classic Platform Release 4.6.0. For detailed requirements, refer to the *AUTOSAR_EXP_CDDDesignAndIntegrationGuideline.pdf*.

1.2 Introduction to CDD

A Complex Driver is a software entity not standardized by AUTOSAR that can access or be accessed via AUTOSAR Interfaces and/or Basic Software Modules APIs.

According to the autosar Layered Software Architecture, a CDD is a specific module located in the Complex Drivers Layer of the Basic Software which interacts with standard BSW modules or Rte.

- A CDD may need to interface to modules of the layered software architecture
- A module of the layered software architecture may need to interface to a CDD
- A CDD may need to interface SW-Cs via RTE



1.3 Hardware Summary

The SENT is a Single Edge Nibble Transmission encoding scheme which conforms to SENT SAE J2716 2010 standard. It is intended for use in applications where high-resolution sensor data needs to be communicated from a sensor to an Electronic Control Unit (ECU). Also, it is used as a replacement for the lower resolution methods of 10-bit A/D's and PWM and as a simpler low-cost alternative to CAN or LIN. The implementation assumes that the sensor is a smart sensor containing a microprocessor or dedicated logic device (ASIC) to create the signal.

In addition to standard protocols, the SENT also supports Short PWM Code (SPC) mode. In this mode, the sensor needs to receive a coordination signal from the controller and then start transmitting the messages. Apart from this mode, the SENT is a unidirectional communications scheme from the sensor to the controller which does not include a coordination signal from the controller. The sensor signal is transmitted as a series of pulses with data encoded as falling to falling edge periods.

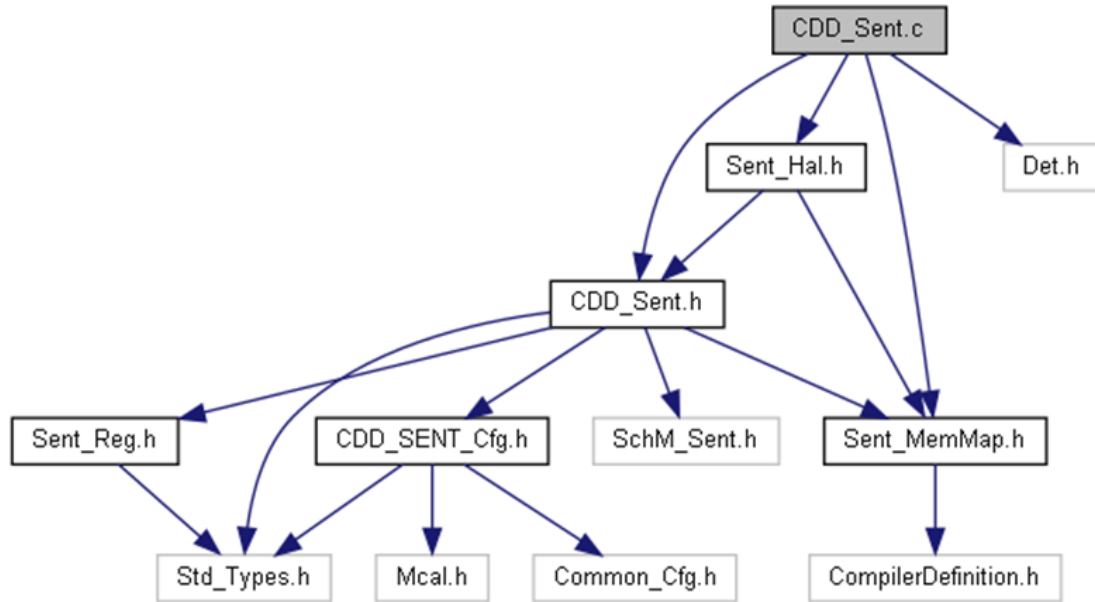
- Support independent receiving of 4-channels data
- Input signal digital filter
- Configurable receive data nibble number
- Configurable receive nibble data mode from H1 to H7
- Calibration pulse compensation function to support clock tick time with a variation of 20%
- Detection of nibble value errors
- Detection of falling edge number errors
- Detection of sync/calibration pulse difference errors
- Detection of calibration diagnosis errors
- Detection of CRC errors
- Two options available for successive calibration pulses diagnosis
- Optional pause pulse in one frame
- Support 76-bit*16 FIFO for each channel
- Support Fast Message, Short Serial Message, and Enhanced Serial Message
- Support SPC mode which supports software triggers or triggers from SOC

Chapter 2 Software Design

2.1 Rejected Requirements

N/A

2.2 File Structure



2.3 Macros

2.3.1 Macros in CDD_Sent.c

- #define **SENT_VENDOR_ID_C** 174
- #define **SENT_AR_RELEASE_MAJOR_VERSION_C** 4
- #define **SENT_AR_RELEASE_MINOR_VERSION_C** 6
- #define **SENT_AR_RELEASE_REVISION_VERSION_C** 0
- #define **SENT_SW_MAJOR_VERSION_C** 0
- #define **SENT_SW_MAJOR_VERSION_C** 4
- #define **SENT_SW_MAJOR_VERSION_C** 0

2.3.2 Macros in CDD_Sent.h

- #define **SENT_VENDOR_ID** 174
- #define **SENT_MODULE_ID** 255

- **#define SENT_AR_RELEASE_MAJOR_VERSION 4**
- **#define SENT_AR_RELEASE_MINOR_VERSION 6**
- **#define SENT_AR_RELEASE_REVISION_VERSION 0**
- **#define SENT_SW_MAJOR_VERSION 0**
- **#define SENT_SW_MINOR_VERSION 4**
- **#define SENT_SW_PATCH_VERSION 0**
- **#define SENT_E_ALREADY_UNINITIALIZED_U8 ((uint8)0x0A)**
API SENT_Sent_DeInit service called when the SENT driver and the Hardware are already uninitialized.
- **#define SENT_E_INVALID_CHANNEL_U8 ((uint8)0x0B)**
API service used with a channel out of range.
- **#define SENT_E_UNINIT_U8 ((uint8)0x0C)**
API service used without module initialization.
- **#define SENT_E_ALREADY_INITIALIZED_U8 ((uint8)0x0D)**
API SENT_Sent_Init service called when the SENT driver and the Hardware are already initialized.
- **#define SENT_E_PARAM_U8 ((uint8)0x0E)**
API service called with wrong parameter.
- **#define SENT_E_INIT_FAILED_U8 ((uint8)0x0FU)**
API Sent_Init service execution failed.
- **#define SENT_E_PARTITION_MAPPING ((uint8)0x10U)**
API service called with wrong core ID.
- **#define SENT_INIT_ID_U8 ((uint8)0x1U)**
API service ID for Sent_Init function.
- **#define SENT_DEINIT_ID_U8 ((uint8)0x2U)**
API service ID for Sent_DeInit function.
- **#define SENT_GETVERSIONINFO_ID_U8 ((uint8)0x3U)**
API service ID for Sent_GetVersionInfo function.

- #define **SENT_REQUEST_SPC_ID_U8** ((uint8)0x4U)
API service ID for Sent_RequestSpcPulse function.
- #define **SENT_FAST_MSG_READ_ID_U8** ((uint8)0x5U)
API service ID for Sent_MainFunctionFastMessageRead function.
- #define **SENT_SLOW_MSG_READ_ID_U8** ((uint8)0x6U)
API service ID for Sent_MainFunctionSlowMessageRead function.

2.4 Enums

2.4.1 Enumerations in CDD_Sent.h

2.4.1.1 Sent_StatusType

Enumeration	Sent_StatusType	
Description	Defines the SENT Status type.	
Values	Value	Description
	SENT_UNINIT = 0x0U	The SENT is not initialized
	SENT_INITIALIZED = 0x1U	The SENT has been initialized

2.4.1.2 Sent_ReturnType

Enumeration	Sent_ReturnType	
Description	Defines the SENT Function Call Result.	
Values	Value	Description
	SENT_OK = 0x0U	Call Correctly
	SENT_NOT_OK = 0x1U	Call Error return
	SENT_BUSY = 0x2U	Transmit request could not be processed because no transmit object was available

2.4.1.3 Sent_DataNibbleModeType

Enumeration	Sent_DataNibbleModeType	
Description	SENT nibble data mode control. Refer to SENT SAE J2716 2010 for detail protocol definition	
Values	Value	Description
	SENT_DATA_NIBBLE_MODE_A = 0x0U	Frame format A
	SENT_DATA_NIBBLE_MODE_H1 = 0x1U	Frame format H1
	SENT_DATA_NIBBLE_MODE_H2 = 0x2U	Frame format H2
	SENT_DATA_NIBBLE_MODE_H3 = 0x3U	Frame format H3

	SENT_DATA_NIBBLE_MODE_H4 0x4U	=	Frame format H4
	SENT_DATA_NIBBLE_MODE_H5 0x5U	=	Frame format H5
	SENT_DATA_NIBBLE_MODE_H6 0x6U	=	Frame format H6
	SENT_DATA_NIBBLE_MODE_H7 0x7U	=	Frame format H7

2.4.1.4 Sent_CalibrationValidType

Enumeration	Sent_CalibrationValidType	
Description	SENT calibration valid type	
Values	Value	Description
	SENT_CALIBRATION_VALID_DISABLE = 0x0U	The successive calibration pulses diagnostic will be disabled
	SENT_CALIBRATION_VALID_WITHIN_20 = 0x1U	The difference of received calibration pulse and receiver configuration is within 20%, the message is deemed to be valid
	SENT_CALIBRATION_VALID_FROM_20_TO_25 = 0x2U	The difference of received calibration pulse and receiver configuration is more than 20% but less than 25%, the message is deemed to be valid

2.4.1.5 Sent_CalDiagOptionType

Enumeration	Sent_CalDiagOptionType	
Description	SENT Successive calibration pulses diagnostic option	
Values	Value	Description
	SENT_CALIBRATION_PULSE_DIAG_OPTION2 = 0x0U	Option 2 i.e. Low Latency Option as per SAE Specification
	SENT_CALIBRATION_PULSE_DIAG_OPTION1 = 0x1U	Option 1 i.e. Preferred but High Latency Option as per SAE Specification

2.4.1.6 Sent_IdleCountType

Enumeration	Sent_IdleCountType	
Description	Determines how long the bus idle flag will assert when SENT bus is idle	
Values	Value	Description
	SENT_IDLE_COUNT_FLAG_DISABLE = 0x0U	The bus idle flag will never assert
	SENT_IDLE_COUNT_FLAG_254_TICKS = 0x1U	The bus is idle for more than 127*2 ticks, then the bus idle flag will assert
	SENT_IDLE_COUNT_FLAG_508_TICKS = 0x2U	The bus is idle for more than 254*2 ticks, then the bus idle flag will assert

	SENT_IDLE_COUNT_FLAG_1016_TICK S = 0x4U	The bus is idle for more than 508*2 ticks, then the bus idle flag will assert
	SENT_IDLE_COUNT_FLAG_2032_TICK S = 0x8U	The bus is idle for more than 1016*2 ticks, then the bus idle flag will assert

2.4.1.7 Sent_SerialMessageType

Enumeration	Sent_SerialMessageType	
Description	SENT Slow message type.	
Values	Value	Description
	SENT_SERIAL_MESSAGE_SHORT = 0x0U	short serial data message
	SENT_SERIAL_MESSAGE_ENHANCE_1 2DATA_8ID = 0x1U	enhanced serial data message with 12-bit data and 8-bit ID
	SENT_SERIAL_MESSAGE_ENHANCE_1 6DATA_4ID = 0x2U	enhanced serial data message with 16-bit data and 4-bit ID

2.4.1.8 Sent_SpcTriggerType

Enumeration	Sent_SpcTriggerType	
Description	SENT trigger type in SPC mode	
Values	Value	Description
	SENT_SPC_SOFTWARE_TRIGGER = 0x0U	SPC pulse triggered by software method
	SENT_SPC_EXTERNAL_TRIGGER = 0x1U	SPC pulse triggered by external trigger

2.4.1.9 Sent_SpcTickBaseType

Enumeration	Sent_SpcTickBaseType	
Description	Select the tick base in SPC mode	
Values	Value	Description
	SENT_SPC_TICK_BASE_PRE_MSG = 0x0U	Previous received message tick base
	SENT_SPC_TICK_BASE_CONFIGURED = 0x1U	SENT configured tick base

2.5 Typedef

2.5.1 Typedefs in CDD_Sent.h

- typedef void (***Sent_FastMsgNotifactionType**)(**Sent_FastMessageDataType *pMsg**)
This type shall be chosen in order to implement fast message error callback function prototype.

- typedef void (***Sent_SlowMsgNotifactionType**)(**Sent_SlowMessageDataType *pMsg**)
This type shall be chosen in order to implement slow message error callback function prototype.

- typedef void (***Sent_FastErrorNotifactionType**)(uint32 u32Status)
This type shall be chosen in order to implement fast message received callback function prototype.
- typedef void (***Sent_SlowErrorNotifactionType**)(uint32 u32Status)
This type shall be chosen in order to implement slow message received callback function prototype.

2.6 Structures

2.6.1 Sent_SpcConfigType

Structure	Sent_SpcConfigType
Description	structure to configure the SENT channel SPC.
Diagram	N/A
Data Fields	• boolean bSpcModeEn Enable SPC mode. • boolean bCalDiagEn Enable Calibration diagnosis. • uint8 u8PulseWidth Configure the width of the SPC pulse. • Uint8 u8PulseDelay Configure the delay between SPC-trigger assert and finally generate a SPC pulse. • Sent_SpcTickBaseType eSpcTickBase Select the tick base of SPC. • Sent_SpcTriggerType eSpcTrigger Select the SPC pulse trigger type.

2.6.2 Sent_ChannelConfigType

Structure	Sent_ChannelConfigType
Description	Structure to configure the SENT channel.
Diagram	N/A
Data Fields	• uint8 u8ChannelLogicId The Channel ID. • uint8 u8ChannelHwId The Channel physical index • uint8 u8DataNibbleNumber The received data number will be configured value plus 1, 0 and value bigger than 5 is forbidden. • uint8 u8DigitalFilterCount The input signal needs keep stable for (filt_cnt<<2)* function_clk cycles to pass through the digital filter. • uint16 u16TickScaler The value plus 1 is used to divide the function clock to tick clock. • boolean bFastMessageFifoEn Enable Channel FIFO function.

- `boolean bTickCompensateEn`
Enable the tick period compensate function.
- `boolean bUseAlternativeCrc`
Use alternative 4-bit CRC algorithm instead of SAE CRC algorithm.
- `boolean bFastMsgCRCCheckEn`
Enable Fast message CRC check.
- `boolean bFastMsgCRCAugEn`
Augmentation is selected for Fast Message CRC calculation.
- `boolean bFastMsgDataChangeEn`
If enabled, the SENT will only assert the fast message ready flag when received data has changed compared with previous data
- `boolean bFastMsgCRCWithSCEn`
If enabled, the crc calculation will includes S&C nibble.
- `boolean bSlowMsgCRCAugEn`
Augmentation is selected for Short Serial Message CRC calculation.
- `boolean bPausePulseEn`
If enabled, a pause pulse will be appended at the end of a frame.
- `boolean bSlowDmaEn`
Enable the Slow message DMA feature.
- `boolean bFastDmaEn`
Enable the Fast message DMA feature.
- `boolean bFastInterruptEn`
Enable the Fast message Interrupt feature.
- `boolean bSlowInterruptEn`
Enable the Slow message Interrupt feature.
- `boolean bChannelEn`
Enable the SENT Channel.
- `Sent_CalDiagOptionType eCalDiagOption`
Successive calibration pulses diagnostic option.
- `Sent_DataNibbleModeType eDataNibbleMode`
Refer to SENT SAE J2716 2010 for detail protocol definition.
- `Sent_CalibrationValidType eCalValid`
Select valid Calibration pulse range
- `Sent_IdleCountType eIdleCount`
Select how long the bus idle flag will assert when bus is idle
- `Sent_FastErrorNotifactionType pFastErrNotification`
Fast message error notification
- `Sent_SlowErrorNotifactionType pSlowErrNotification`
Slow message error notification
- `Sent_FastMsgNotifactionType pFastMsgNotification`
Fast message Receive notification
- `Sent_SlowMsgNotifactionType pSlowMsgNotification`
Slow message Receive notification
- `Sent_SpcConfigTypep *pSpcModeCfg`

The Pointer of the Sent_SpcConfigType

2.6.3 Sent_InstanceConfigType

Structure	Sent_InstanceConfigType
Description	Structure to configure the SENT instance.
Diagram	N/A
Data Fields	• uint8 u8InstanceLogicIndex The controller ID • uint8 u8InstanceHwIndex The controller physical index • boolean bControllerEn Sent Controller Activation • boolean bFifoOverflowEn Fast message FIFO overflow interrupt enable • boolean bDebugModeEn If enable, the SENT will keep running under debug mode. • boolean bAutoClearReadyFlag When enabled after data ready read of received data, crc and timestamp register will prevent the assert of data overflow flag. • uint8 u8WaterMark When Fast message DMA request is enabled, the request will assert if data number in FIFO is more than watermark • uint8 u8Prescaler This value is used to divide the function clock to count the timestamp • uint8 u8ChannelCount The counter of channel used • Sent_ChannelConfigType **pSentChannelCfg The Pointer array of the Sent_ChannelConfigType

2.6.4 Sent_ConfigType

Structure	Sent_ConfigType
Description	Data structure containing the set of configuration parameters required for initializing the Sent instance(s) and channel(s).
Diagram	N/A
Data Fields	• uint8 u8InstanceCount The counter of the used Sent instance • Sent_InstanceConfigType **pSentInstanceCfg The Pointer array of the Sent_InstanceConfigType • Boolean *Sent_CoresMappingPtr The Pointer array of the core mapping, if true, means the current core need to manage at least one sent instance • uint32 *Sent_CtrlCoresMappingPtr

	The Pointer array of the controller mapping, the value indicate which core should manage the sent instance
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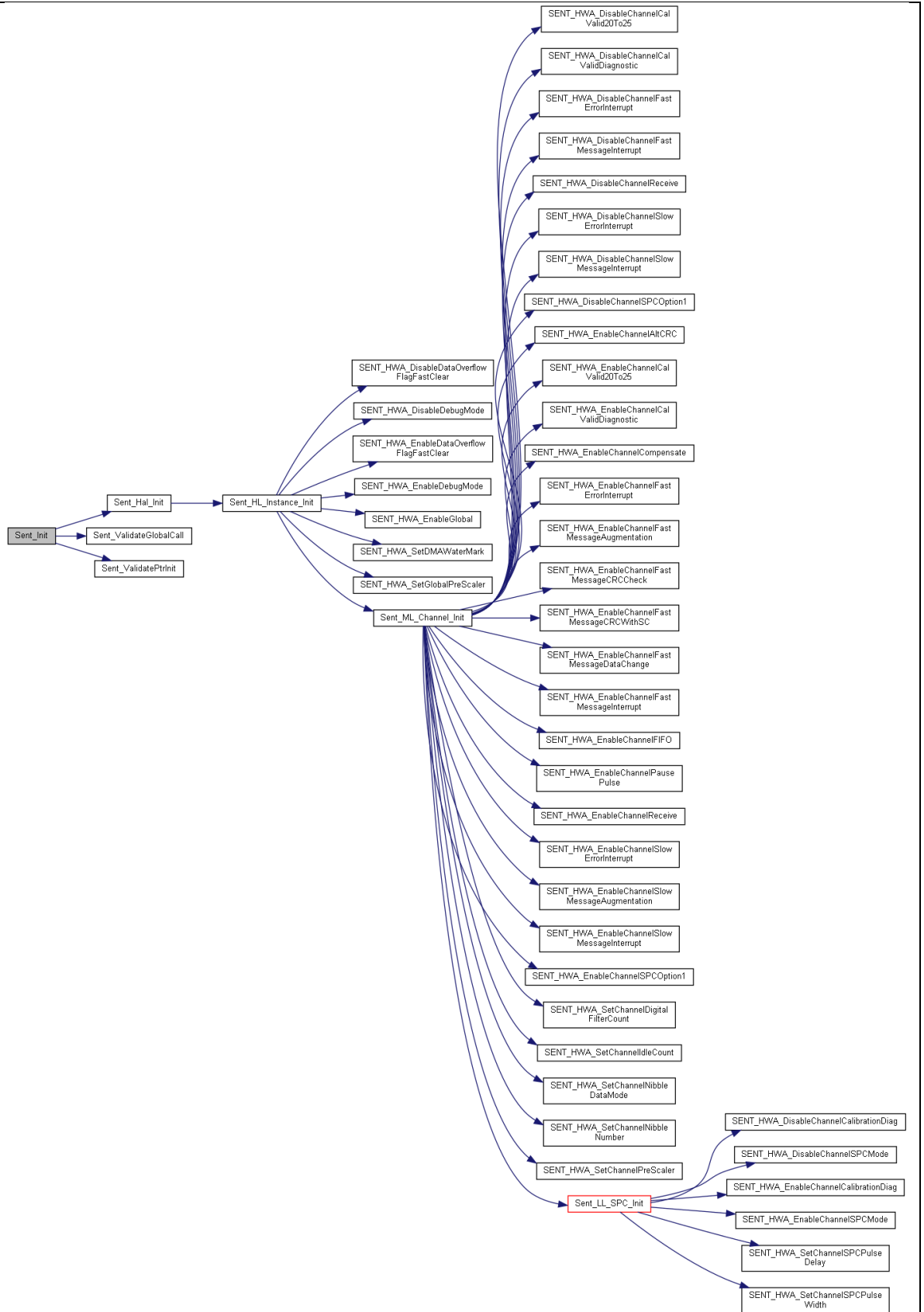
2.7 API Functions

2.7.1 Functions in Cdd_Sent.c

2.7.1.1 void Sent_Init(const Sent_ConfigType * pConfigPtr)

Function	void Sent_Init(const Sent_ConfigType * pConfigPtr)
Description	Initialize the sent instance configuration.

Diagram



Parameters

Parameter

Description

pConfigPtr

The pointer of sent instance configuration structure

Returns

void

2.7.1.2 void Sent_DeInit(void)

Function

void Sent_DeInit(void)

Description	This function de-initializes the SENT driver. This service is a non-reentrant function. Returns all underlying hardware to a state comparable to their power on reset state, and de-initialize the SENT driver. This service is a non-reentrant function. Returns all underlying hardware to a state comparable to their power on reset state, and de-initialize the SENT driver.	
Diagram	<pre> graph LR Sent_DeInit --> Sent_Hal_DeInit Sent_Hal_DeInit --> Sent_HL_Instance_DeInit Sent_HL_Instance_DeInit --> SENT_HWA_DisableGlobal Sent_HL_Instance_DeInit --> Sent_LL_Channel_DeInit Sent_LL_Channel_DeInit --> SENT_HWA_ClearChannelIntStatus Sent_LL_Channel_DeInit --> SENT_HWA_DisableChannelReceive Sent_LL_Channel_DeInit --> SENT_HWA_SetChannelCtrlReg </pre>	
Parameters	Parameter	Description
	N/A	N/A
Returns	void	

2.7.1.3 void Sent_GetVersionInfo(Std_VersionInfoType * pVersioninfo)

Function	void Sent_GetVersionInfo(Std_VersionInfoType * pVersioninfo)	
Description	This service returns the version information of this module. This service is Non-reentrant and returns the version information of this module. The version information includes: - Module Id - Vendor Id - Vendor specific version numbers. If source code for caller and callee of this function is available this function should be realized as a macro. The macro should be defined in the modules header file.	
Diagram	<pre> graph LR Sent_GetVersionInfo --> Sent_ValidateCommonPointer </pre>	
Parameters	Parameter	Description
	pVersioninfo	Pointer to where to store the version information of this module.
Returns	N/A	

2.7.1.4 void Sent_RequestSpcPulse (uint8 u8ControllerID,uint8 u8ChannelID)

Function	void Sent_RequestSpcPulse(uint8 u8ControllerID,uint8 u8ChannelID)	
Description	This service request the SPC pulse. The transmission occurs depending on an SPC Pulse (a synchronization signal) from the receiver module, and the SPC Pulse is implemented as a low pulse. When the sensor receives the coordination signal from SOC and then it starts transmitting the messages.	
Diagram	<pre> graph LR Sent_RequestSpcPulse --> SENT_Hal_RequestSPCPulse Sent_RequestSpcPulse --> Sent_ValidateControllerChannelIndex Sent_RequestSpcPulse --> Sent_ValidateGlobalCall SENT_Hal_RequestSPCPulse --> Sent_AB_GetHwSentChannel SENT_Hal_RequestSPCPulse --> SENT_HWA_StartChannelSPC Sent_ValidateControllerChannelIndex --> Sent_ValidateInitStatus </pre>	

Parameters	Parameter	Description
	u8ControllerID	The sent controller index
	u8ChannelID	The sent channel index
Returns	void	

2.7.1.5 void Sent_MainFunctionFastMessageRead(void)

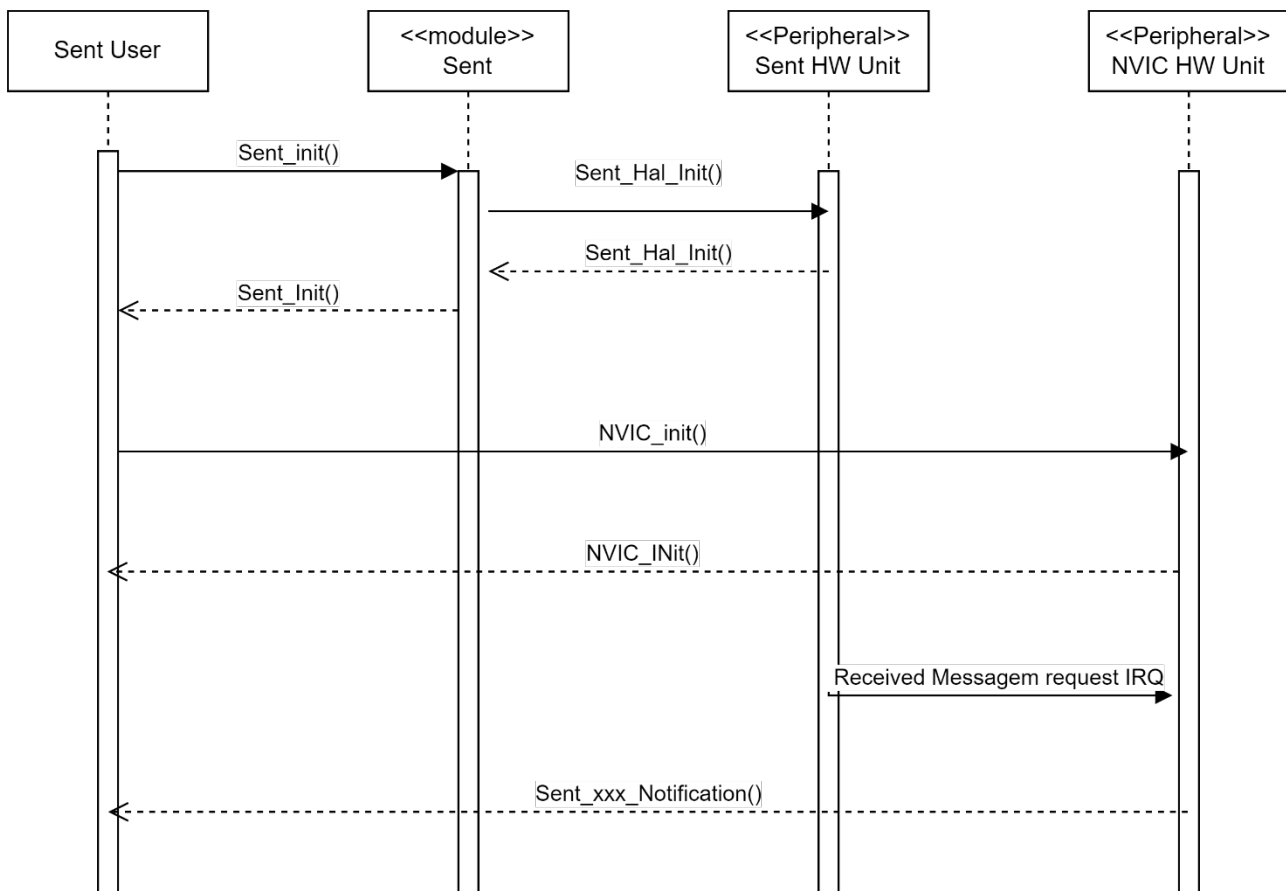
Function	void Sent_MainFunctionFastMessageRead(void)	
Description	This function performs the polling of reading Fast message when FastMsgReadType is set to Polling.	
Diagram	<pre> graph LR Sent_MainFunctionFastMessageRead --> SENT_Hal_MainFunctionFastMsgRead Sent_MainFunctionFastMessageRead --> Sent_ValidateInitStatus SENT_Hal_MainFunctionFastMsgRead --> Sent_HL_GetFastMessage Sent_HL_GetFastMessage --> Sent_AB_GetHwSentChannel Sent_HL_GetFastMessage --> SENT_HWA_GetChannelDataNibble Sent_HL_GetFastMessage --> SENT_HWA_GetChannelFastMessageCRCNibble Sent_HL_GetFastMessage --> SENT_HWA_GetChannelFastMessageStatusNibble Sent_HL_GetFastMessage --> SENT_HWA_GetChannelFastMessageTimeStamp Sent_HL_GetFastMessage --> SENT_HWA_GetChannelNibbleNumber Sent_HL_GetFastMessage --> SENT_HWA_GetFastMessageReadyFlag </pre>	
Parameters	Parameter	Description
	N/A	N/A
Returns	void	

2.7.1.6 void Sent_MainFunctionSlowMessageRead(void)

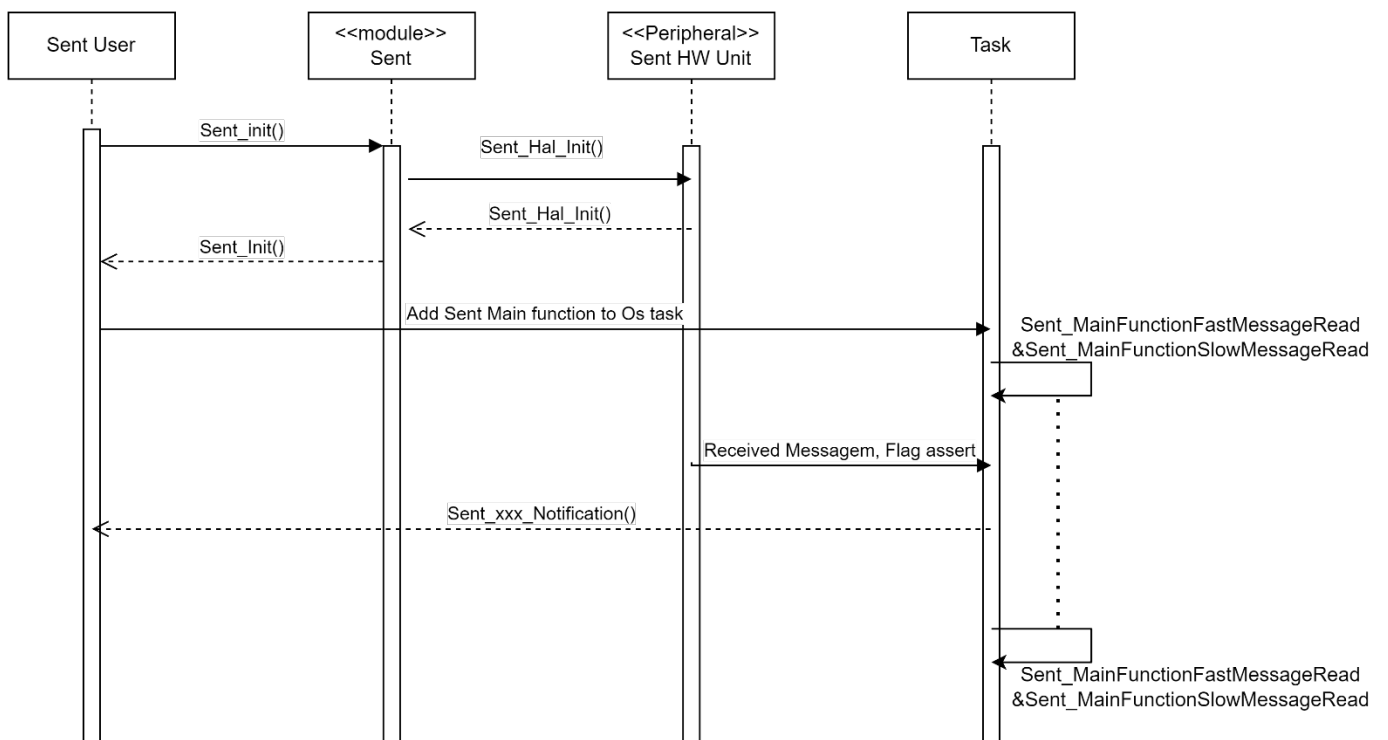
Function	void Sent_MainFunctionSlowMessageRead(void)	
Description	This function performs the polling of reading Slow message when SlowMsgReadType is set to Polling.	
Diagram	<pre> graph LR Sent_MainFunctionSlowMessageRead --> SENT_Hal_MainFunctionSlowMsgRead Sent_MainFunctionSlowMessageRead --> Sent_ValidateInitStatus SENT_Hal_MainFunctionSlowMsgRead --> Sent_HL_GetSlowMessage Sent_HL_GetSlowMessage --> Sent_AB_GetHwSentChannel Sent_HL_GetSlowMessage --> SENT_HWA_GetChannelBit2CRC Sent_HL_GetSlowMessage --> SENT_HWA_GetChannelBit2DATA Sent_HL_GetSlowMessage --> SENT_HWA_GetChannelBit3EnhancedID3_0_OR_DATA15_12 Sent_HL_GetSlowMessage --> SENT_HWA_GetChannelBit3EnhancedID7_4_OR_ID3_0 Sent_HL_GetSlowMessage --> SENT_HWA_GetChannelSlowMessageTimeStamp Sent_HL_GetSlowMessage --> SENT_HWA_GetChannelSlowMessageType Sent_HL_GetSlowMessage --> SENT_HWA_GetSlowMessageReadyFlag </pre>	
Parameters	Parameter	Description
	N/A	N/A
Returns	void	

2.8 API Sequence Diagram

2.8.1 Sent Initialization with Interrupt Mode



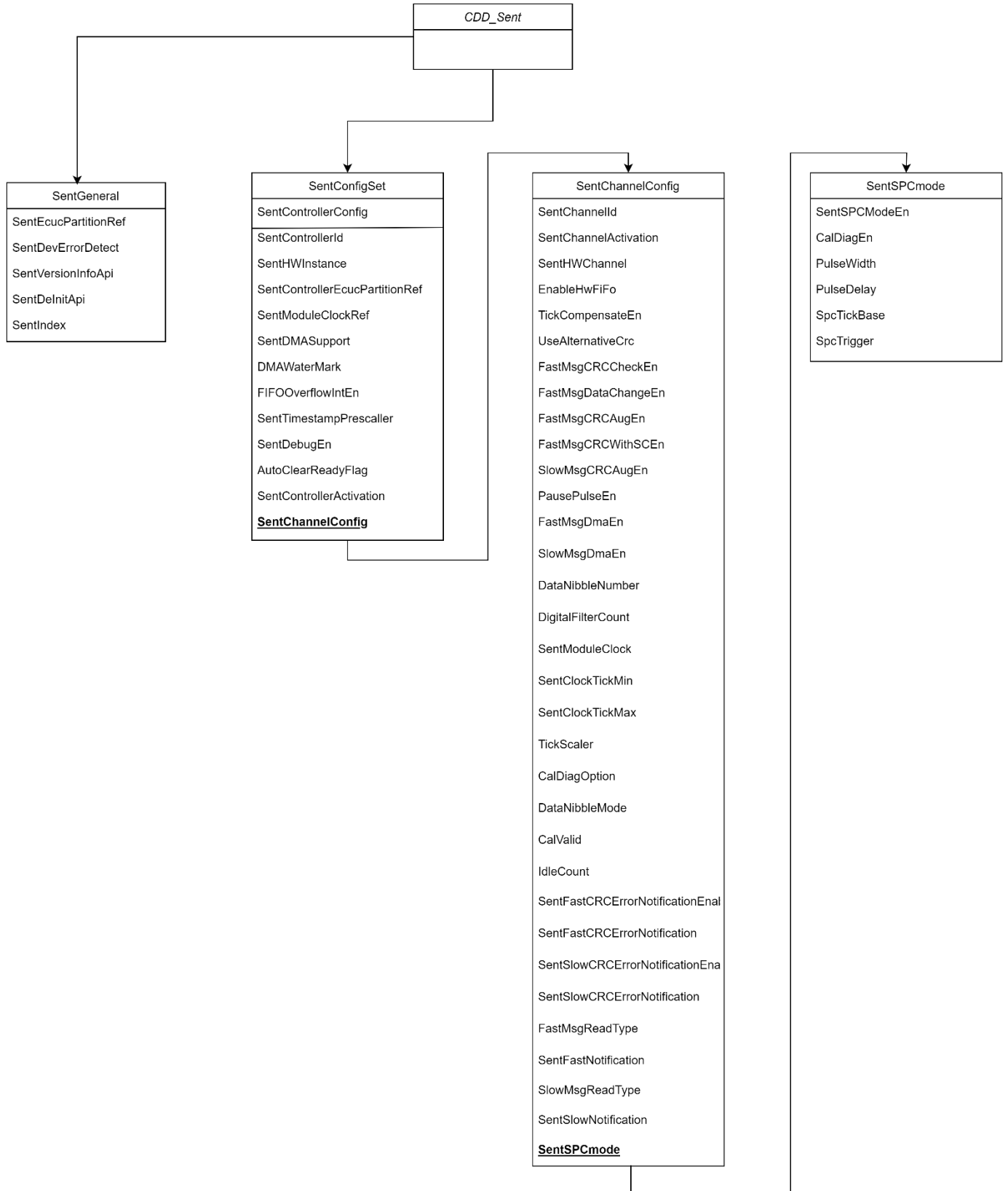
2.8.2 Sent Initialization with Polling Mode



Chapter 3 Tresos Configuration Items

3.1 Container Inclusion Relation

The container inclusion relation is shown as below:

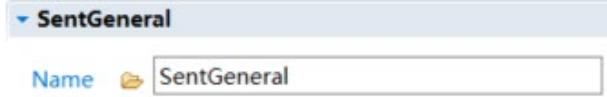


3.2 Containers and Variables

3.2.1 IMPLEMENTATION_CONFIG_VARIANT

Container	IMPLEMENTATION_CONFIG_VARIANT	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	ENUMERATION
	Label	Config Variant
	Default	VariantPreCompile

3.2.2 SentGeneral

Container	SentGeneral	
Description	Vendor specific: Configuration of general Sent parameters.	
Screenshot		
Properties	Property	Value
	Type	IDENTIFIABLE
	Label	N/A
	Default	N/A

3.2.2.1 DmaDevErrorDetect

Container	SentDevErrorDetect	
Description	Switches the Development Error Detection and Notification on or off. - true: Enabled. - false: Disabled.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	SentDev Error Detect
	Default	false

3.2.2.2 SentMulticoreSupport

Container	SentMulticoreSupport	
Description	Switches multicore support on or off: False: For all variants, no EcucPartition shall be referenced in SentEcucPartitionRef. True: For all variants, at least one EcucPartition needs to be referenced in SentEcucPartitionRef.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	SentMulticoreSupport
	Default	false

3.2.2.3 SentDelInitApi

Container	SentDelInitApi	
Description	Vendor specific: Enables/Disables the Sent_DeInit API.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Provide De-Init API
	Default	false

3.2.2.4 SentVersionInfoApi

Container	SentVersionInfoApi	
Description	Switches the Sent_GetVersionInfo() API: ON or OFF. When this option is ON driver supports API for getting Version information for the Driver.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Provide Version Info API
	Default	false

3.2.2.5 SentIndex

Container	SentIndex	
Description	Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Label	Sent Driver Index
	Default	0

3.2.3 SentGeneral

Container	SentGeneral	
Description	This container holds the parameters related each SENT Driver Unit.	
Screenshot		
Properties	Property	Value
	Type	IDENTIFIABLE
	Label	N/A
	Default	N/A

3.2.3.1 SentControllerId

Container	SentControllerId	
Description	This parameter provides the controller ID which is unique in a given SENT Driver. The value for this parameter starts with 0 and continue without any gaps.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Label	Sent Controller ID
	Default	0->255

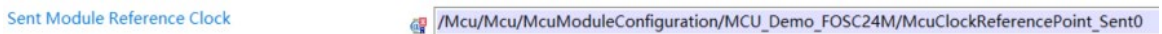
3.2.3.2 SentHWInstance

Container	SentHWInstance	
Description	Specifies which one of the on-chip SENT interfaces is associated with this controller ID.	
Screenshot		
Properties	Property	Value
	Type	STRING(RANGE)
	Label	Sent hardware Instance
	Default	SENT0

3.2.3.3 SentControllerEcucPartitionRef

Container	SentControllerEcucPartitionRef	
Description	EN: Maps a SENT controller to zero or multiple ECUC partitions to limit the access to this controller. The ECUC partitions referenced are a subset of the ECUC partitions where the SENT driver is mapped to.	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Label	SentControllerEcucPartitionRef
	Default	N/A

3.2.3.4 SentModuleClockRef

Container	SentModuleClockRef	
Description	Reference to the CPU clock configuration, which is set in the MCU driver configuration. MCU plugin need to be added and then give the reference to it.	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Label	Sent Module Reference Clock
	Default	N/A

3.2.3.5 SentDMASupport

Container	SentDMASupport	
------------------	----------------	--

Description	The logic switch for DMA support	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Sent DMA Support
	Default	false

3.2.3.6 DMAWaterMark

Container	DMAWaterMark	
Description	The DMA watermark should work with fast message fifo and DMA support. When Fast message DMA request is enabled, the request will asserts if data number in FIFO is more than watermark.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Label	DMA Watermark
	Default	0

3.2.3.7 FIFOOverflowIntEn

Container	FIFOOverflowIntEn	
Description	Global control register of fast message FIFO overflow interrupt generate. The interrupt only can be enabled when sent DMA support.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Fast message FIFO overflow interrupt enable
	Default	false

3.2.3.8 SentTimestampPrescaler

Container	SentTimestampPrescaler	
Description	0 to 255 - Time Stamp Clock = (High Frequency Clock)/(SentTimestampPrescaler + 1). Value written in this field should be the value of SentTimestampPrescaler from above formula. Value set by user software should be such that the clock period is at least 1 μ s.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Label	Time Stamp Prescaler
	Default	0

3.2.3.9 SentDebugEn

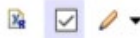
Container	SentDebugEn	
Description	When this parameter is enabled, SENT will stop running under debug mode	

Screenshot	Sent Debug Mode Enable 	
Properties	Property	Value
	Type	BOOLEAN
	Label	Sent Debug Mode Enable
	Default	false

3.2.3.10 AutoClearReadyFlag

Container	AutoClearReadyFlag	
Description	When enabled after data ready read of received data, crc and timestamp register will prevent the assert of data overflow flag.	
Screenshot	Auto Clear Ready Flag 	
Properties	Property	Value
	Type	BOOLEAN
	Label	Auto Clear Ready Flag
	Default	false

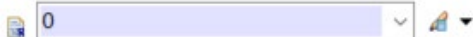
3.2.3.11 SentControllerActivation

Container	SentControllerActivation	
Description	Defines if a SENT controller is used in the configuration. Deactivation of a particular SENT controller is equivalent to a SENT controller not used in the configuration.	
Screenshot	Sent Controller Activation 	
Properties	Property	Value
	Type	BOOLEAN
	Label	Sent Controller Activation
	Default	true

3.2.4 SentChannelConfig

Container	SentChannelConfig	
Description	This container contains the configuration (parameters) of the SENT channels.	
Screenshot	 SentChannelConfig	
Properties	Property	Value
	Type	MAP
	Label	N/A
	Default	N/A

3.2.4.1 SentChannelId

Container	SentChannelId	
Description	This parameter provides the Channel ID which is unique in a given SENT channel. The value for this parameter starts with 0 and continue without any gaps.	
Screenshot	Sent Channel Id 	

Properties	Property	Value
	Type	INTEGER
	Label	Sent Channel Id
	Default	0

3.2.4.2 SentChannelActivation

Container	SentChannelActivation	
Description	Defines if a SENT Channel is used in the configuration. Deactivation of a particular SENT Channel is equivalent to a SENT Channel not used in the configuration.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Sent Channel Activation
	Default	true

3.2.4.3 SentHWChannel

Container	SentHWChannel	
Description	Specifies which one of the on-chip SENT interfaces is associated with this controller ID.	
Screenshot		
Properties	Property	Value
	Type	STRING(RANGE)
	Label	Sent hardware Channel
	Default	Sent_Channel0

3.2.4.4 EnableHwFiFo

Container	EnableHwFiFo	
Description	Enable or disable hardware FIFO for this controller. If DMA is not enabled, this node is unused	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Enable Sent Controller hardware FiFo(fast message)
	Default	false

3.2.4.5 TickCompensateEn

Container	TickCompensateEn	
Description	Enable the tick period compensate function.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Enable tick period compensate

	Default	false
--	---------	-------

3.2.4.6 UseAlternativeCrc

Container	UseAlternativeCrc	
Description	Enable the tick period compensate function.	
Screenshot	Use alternative 4-bit CRC algorithm 	
Properties	Property	Value
	Type	BOOLEAN
	Label	Use alternative 4-bit CRC algorithm
	Default	false

3.2.4.7 FastMsgCRCCheckEn

Container	FastMsgCRCCheckEn	
Description	Enable Fast message CRC check	
Screenshot	Enable Fast Message CRC check 	
Properties	Property	Value
	Type	BOOLEAN
	Label	Enable Fast Message CRC check
	Default	false

3.2.4.8 FastMsgDataChangeEn

Container	FastMsgDataChangeEn	
Description	Enable Fast message CRC check	
Screenshot	Enable Fast Message Data change check 	
Properties	Property	Value
	Type	BOOLEAN
	Label	Enable Fast Message Data change check
	Default	false

3.2.4.9 FastMsgCRCAugEn

Container	FastMsgCRCAugEn	
Description	calculation.	
Screenshot	Add augmentation into the Fast Message CRC calculation 	
Properties	Property	Value
	Type	BOOLEAN
	Label	Add augmentation into the Fast Message CRC calculation
	Default	false

3.2.4.10 FastMsgCRCWithSCEn

Container	FastMsgCRCWithSCEn	
Description	If enabled, the crc calculation will includes S&C nibble	

Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Include SC nibble for Fast Message CRC calculation
	Default	false

3.2.4.11 SlowMsgCRCAugEn

Container	SlowMsgCRCAugEn	
Description	select whether the augmentation is added into the CRC calculation. If selected, a 4-bit zero will be added at the end of the CRC calculation.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Add augmentation into the Slow Message CRC calculation
	Default	false

3.2.4.12 PausePulseEn

Container	PausePulseEn	
Description	If enabled, a pause pulse will be appended at the end of a frame.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Append Pause Pulse at end of frame
	Default	false

3.2.4.13 FastMsgDmaEn

Container	FastMsgDmaEn	
Description	Enable the fast message DMA function.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Enable Fast Message DMA
	Default	false

3.2.4.14 SlowMsgDmaEn

Container	SlowMsgDmaEn	
Description	Enable the slow message DMA function.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Enable Fast Message DMA

	Default	false
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3.2.4.15 DataNibbleNumber

Container	DataNibbleNumber	
Description	2 to 6 Data Nibbles supported in this Channel	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Label	Data Nibble Number
	Default	6

3.2.4.16 DigitalFilterCount

Container	DigitalFilterCount	
Description	The input signal needs keep stable for (filt_cnt<<2)* function_clk cycles to pass through the digital filter.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Label	Digital Filter Count
	Default	0

3.2.4.17 SentModuleClock

Container	SentModuleClock	
Description	The frequency of Sent clock source.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Label	Sent Module Clock (Hz) - Reselect after clock change
	Default	0

3.2.4.18 SentClockTickMin

Container	SentClockTickMin	
Description	Specifies the Min Clock Tick of the sent in us.	
Screenshot		
Properties	Property	Value
	Type	FLOAT
	Label	Sent Clock Tick Min Value(us)
	Default	2.0

3.2.4.19 SentClockTickMax

Container	SentClockTickMax	
Description	Specifies the Max Clock Tick of the sent in us.	

Screenshot	Sent Clock Tick MAn Value(us) (0 -> 5000) 	
Properties	Property	Value
	Type	FLOAT
	Label	Sent Clock Tick Max Value(us)
	Default	3.88

3.2.4.20 TickScaler

Container	TickScaler	
Description	The value is used to divide the function clock to tick clock.	
Screenshot	Channel Tick Scaler (0 -> 255) 	
Properties	Property	Value
	Type	INTEGER
	Label	Channel Tick Scaler
	Default	0

3.2.4.21 CalDiagOption

Container	CalDiagOption	
Description	Successive calibration pulses diagnostic option Option 2: i.e. Low Latency Option as per SAE Specification Option 1: i.e. Preferred but High Latency Option as per SAE Specification	
Screenshot	Successive calibration pulses diagnostic option* 	
Properties	Property	Value
	Type	STRING(RANGE)
	Label	Successive calibration pulses diagnostic option
	Default	SENT_CALIBRATION_PULSE_DIAG_OPTION1

3.2.4.22 DataNibbleMode

Container	DataNibbleMode	
Description	Refer to SENT SAE J2716 2010 for detail protocol definition.	
Screenshot	Nibble Data Mode 	
Properties	Property	Value
	Type	STRING(RANGE)
	Label	Nibble Data Mode
	Default	SENT_DATA_NIBBLE_MODE_A

3.2.4.23 CalValid

Container	CalValid	
Description	Select valid Calibration pulse range.	
Screenshot	valid Calibration pulse range* 	
Properties	Property	Value
	Type	STRING(RANGE)

	Label	valid Calibration pulse range
	Default	SENT_CALIBRATION_VALID_DISABLE

3.2.4.24 IdleCount

Container	IdleCount	
Description	This field determines how long the bus idle flag will assert when bus is idle.	
Screenshot		
Properties	Property	Value
	Type	STRING(RANGE)
	Label	Idle time interval
	Default	SENT_IDLE_COUNT_FLAG_DISABLE

3.2.4.25 SentFastCRCErrNotificationEnable

Container	SentFastCRCErrNotificationEnable	
Description	Enable Fast Error Notification.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Sent Fast Error Notification Enable
	Default	false

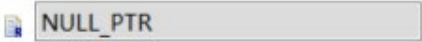
3.2.4.26 SentFastCRCErrNotification

Container	SentFastCRCErrNotification	
Description	This parameter defines the existence and the name of a callout function that is called after an error occurred. No call should occur if this is not enabled.	
Screenshot		
Properties	Property	Value
	Type	STRING
	Label	Sent Fast Error Notification
	Default	NULL_PTR

3.2.4.27 SentSlowCRCErrNotificationEnable

Container	SentSlowCRCErrNotificationEnable	
Description	Enable Slow Error Notification.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Sent Slow Error Notification Enable
	Default	false

3.2.4.28 SentSlowCRCErrNotification

Container	3.2.4.28 SentSlowCRCErrNotification	
Description	This parameter defines the existence and the name of a callout function that is called after an error occurred. No call should occur if this is not enabled.	
Screenshot		
Properties	Property	Value
	Type	STRING
	Label	Sent Slow Error Notification
	Default	NULL_PTR

3.2.4.29 FastMsgReadType

Container	FastMsgReadType	
Description	define the how to read fast message. Interrupt or polling.	
Screenshot		
Properties	Property	Value
	Type	STRING(RANGE)
	Label	Fast Message Read type
	Default	Interrupt

3.2.4.30 SentFastNotification

Container	SentFastNotification	
Description	This parameter defines the existence and the name of a callout function that is called after a fast message is received.	
Screenshot		
Properties	Property	Value
	Type	STRING
	Label	Sent Fast Notification
	Default	SentFastNotification

3.2.4.31 SlowMsgReadType

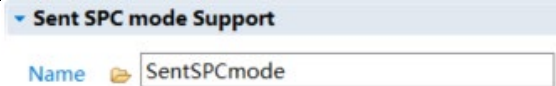
Container	SlowMsgReadType	
Description	define the how to read slow message. Interrupt or polling.	
Screenshot		
Properties	Property	Value
	Type	STRING(RANGE)
	Label	Slow Message Read type
	Default	Interrupt

3.2.4.32 SentSlowNotification

Container	SentSlowNotification
------------------	----------------------

Description	This parameter defines the existence and the name of a callout function that is called after a slow message is received.	
Screenshot		
Properties	Property	Value
	Type	STRING
	Label	Sent Slow Notification
	Default	SentSlowNotification

3.2.5 SentSPCmode

Container	SentSPCmode	
Description	This container contains the configuration (parameters) of the Sent SPC mode.	
Screenshot		
Properties	Property	Value
	Type	Container
	Label	Sent SPC mode Support
	Default	N/A

3.2.5.1 SentSPCModeEn

Container	SentSPCModeEn	
Description	Defines if SPC mode is used in the sent configuration.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Sent SPC Mode Enable
	Default	false

3.2.5.2 CalDiagEn

Container	CalDiagEn	
Description	Enable SPC mode Calibration diagnosis.	
Screenshot		
Properties	Property	Value
	Type	BOOLEAN
	Label	Sent SPC Mode Calibration diagnosis Enable
	Default	false

3.2.5.3 PulseWidth

Container	PulseWidth	
Description	Configure the width of the SPC pulse.	
Screenshot		
Properties	Property	Value

	Type	INTEGER
	Label	Width Of The SPC Pulse
	Default	1

3.2.5.4 PulseDelay

Container	PulseDelay	
Description	Configure the delay between SPC-trigger assert and finally generate a SPC pulse.	
Screenshot		
Properties	Property	Value
	Type	INTEGER
	Label	Delay Of The SPC Pulse
	Default	1

3.2.5.5 SpcTickBase

Container	SpcTickBase	
Description	Select the tick base of SPC	
Screenshot		
Properties	Property	Value
	Type	STRING(RANGE)
	Label	Tick base of SPC
	Default	SENT_SPC_TICK_BASE_CONFIGURED

3.2.5.6 SpcTrigger

Container	temp SpcTrigger	
Description	Select the SPC pulse trigger type.	
Screenshot		
Properties	Property	Value
	Type	STRING(RANGE)
	Label	SPC pulse trigger type
	Default	SENT_SPC_EXTERNAL_TRIGGER

3.2.6 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	Property	Value
	Type	IDENTIFIABLE
	Label	N/A

	Default	N/A
--	---------	-----

3.2.6.1 AUTOSAR Major Version

Container	AUTOSAR Major Version	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot	AUTOSAR Major Version  4	
Properties	Property	Value
	Type	INTEGER_LABEL
	Label	N/A
	Default	4

3.2.6.2 AUTOSAR Minor Version

Container	AUTOSAR Minor Version	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot	AUTOSAR Minor Version  6	
Properties	Property	Value
	Type	INTEGER_LABEL
	Label	N/A
	Default	6

3.2.6.3 AUTOSAR Patch Version

Container	AUTOSAR Patch Version	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot	AUTOSAR Patch Version  0	
Properties	Property	Value
	Type	INTEGER_LABEL
	Label	N/A
	Default	0

3.2.6.4 Software Major Version

Container	Software Major Version	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot	Software Major Version  0	
Properties	Property	Value
	Type	INTEGER_LABEL
	Label	N/A
	Default	0

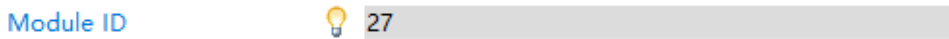
3.2.6.5 Software Minor Version

Container	Software Minor Version	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	Label	N/A
	Default	3

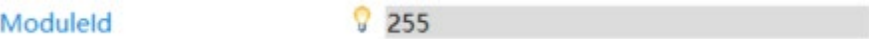
3.2.6.6 Software Patch Version

Container	Software Patch Version	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	Label	N/A
	Default	0

3.2.6.7 Module ID

Container	Module ID	
Description	Module ID of this module from Module List.	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	Label	N/A
	Default	255

3.2.6.8 Vendor ID

Container	Vendor ID	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	Label	N/A
	Default	174

Chapter 4 Configuration Guides

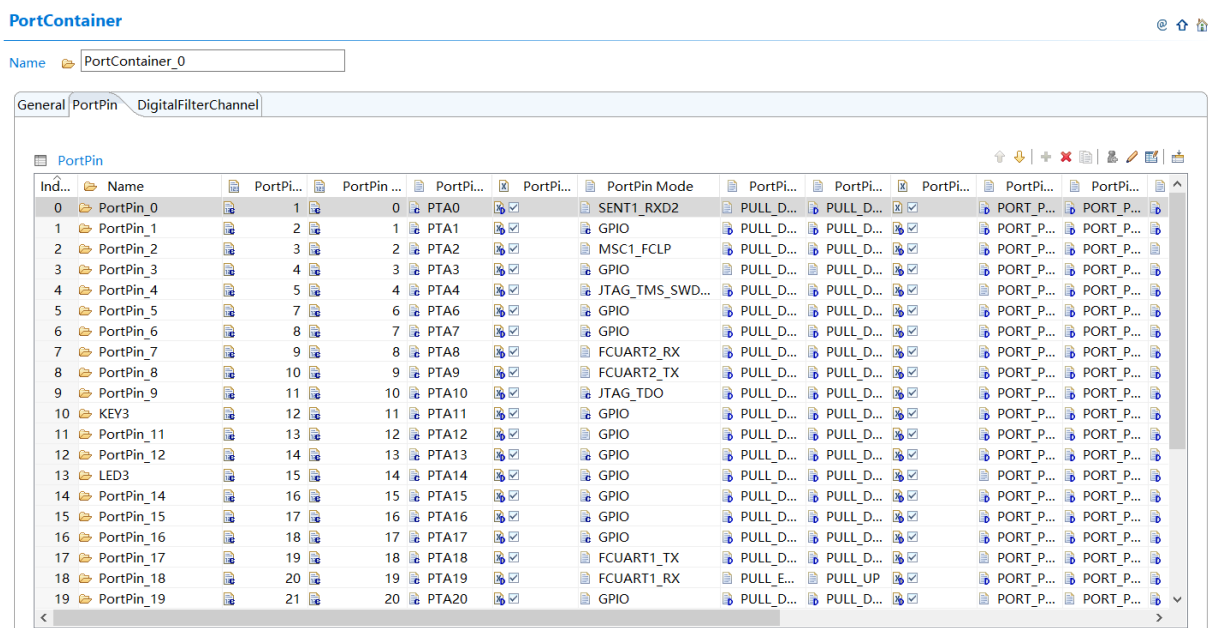
4.1 SENT Usage Common Steps

Basically, the Sent module can be configured in 4 steps:

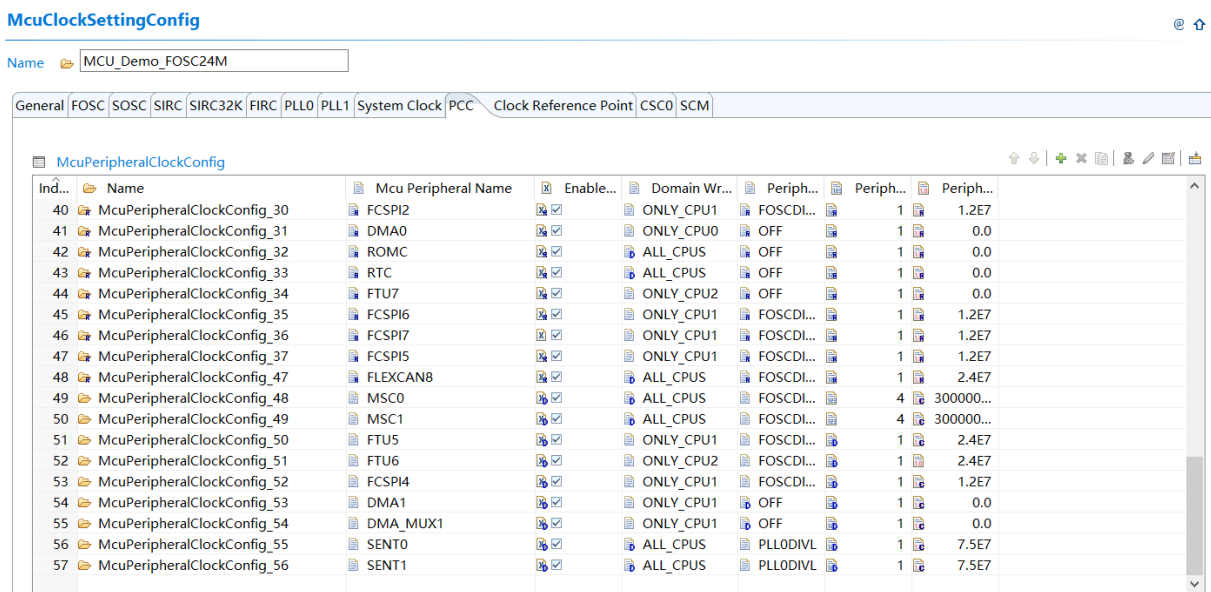
- Configure Sent general configurations and enable Sent Port&clock configuration.
- Configure Sent controller configuration.
- Configure Sent channel configuration
- Generate configuration files.

4.2 Sent Usage Demo

- 1) Users should set the Sent pin mode in PortContainer firstly.



- 2) Then set the Sent clock in McuClockReferencepoint_PCC container.



- 3) In Sent General configuration tag, users can configure if SentMulticoreSupport is used. If Det isn't needed, the Development Error Detection checkbox has no need to be checked.

SENT

Name

SENT

General

SentControllerConfig

SentEcucPartitionRef

Published Information

Config Variant

VariantPreCompile

▼ SentConfigSet

Name

SentConfigSet

▼ SentGeneral

Name

SentGeneral

Development Error Detection

SentMulticoreSupport

Provide Version Info API

Provide De-Init API

Sent Driver Index (0 -> 255)

0

- 4) In SentEcucPartitionRef tag, users can select what cores would use the sent controller.

SENT

Name

SENT

General

SentControllerConfig

SentEcucPartitionRef

Published Information

SentEcucPartitionRef

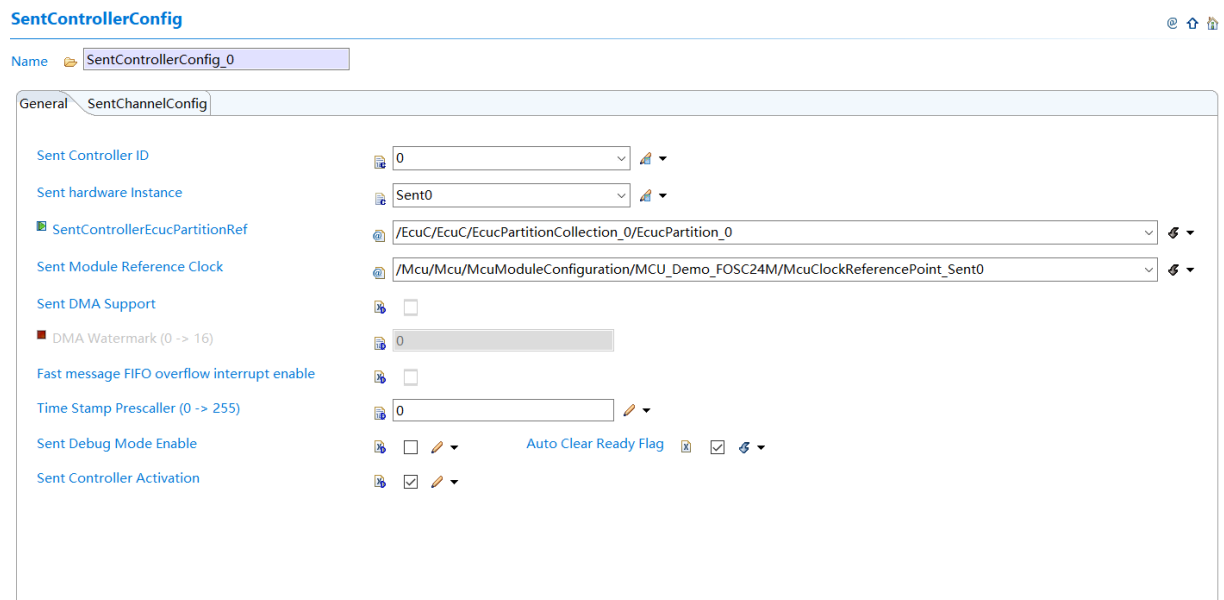
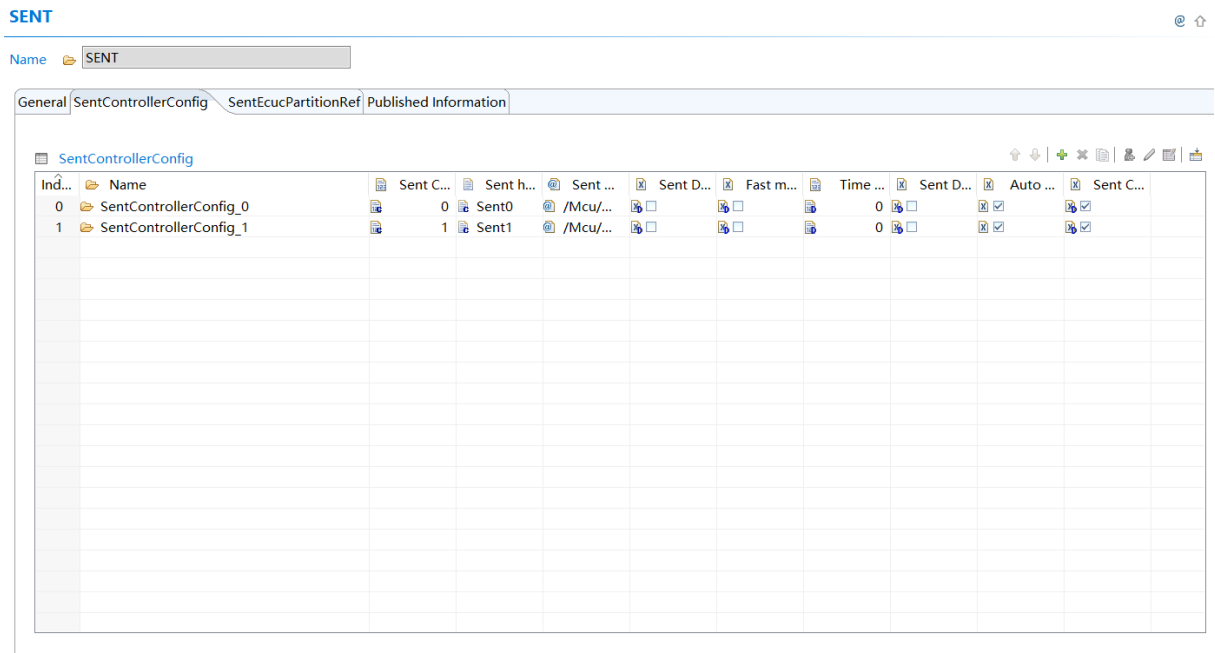
Ind...

SentEcucPartitionRef

0

/EcuC/EcuC/EcucPartitionCollection_0/EcucPartition_0

- 5) In SentControllerConfig tag, users can select what sent controller would be configured and used.



- 6) In SentChannelConfig tag, users can select what sent channel would be configured and used.

SentChannelConfig

@

Name SentChannelConfig_MLX90367LDC

General

Sent Channel Id	0		
Sent Channel Activation	☒		
Sent hardware Channel	Sent_Channel2		
Enable Sent Controller hardware FiFo(fast message)	☐	Enable tick period compensate	☒
Use alternative 4-bit CRC algorithm	☐	Enable Fast Message CRC check	☒
Enable Fast Message Data change check	☐	Add augmentation into the Fast Message CRC calculation	☒
Include SC nibble for Fast Message CRC calculation	☐	Add augmentation into the Slow Message CRC calculation	☒
Append Pause Pulse at end of frame	☒	Enable Fast Message DMA	☐
Enable Slow Message DMA	☐		
Data Nibble Number (2 -> 6)	6		
Digital Filter Count (0 -> 255)	4		
Sent Module Clock (Hz) - Reselect after clock change	75000000		
Sent Clock Tick Min Value(us) (0 -> 5000)	2.0		
Sent Clock Tick MAn Value(us) (0 -> 5000)	3.88		
Channel Tick Scaler (0 -> 255)	220		
Successive calibration pulses diagnostic option	-IBRATION_PULSE_DIAG_OPTION1		

SentChannelConfig

Name SentChannelConfig_MLX90367LDC

General

Idle time interval	SENT_IDLE_COUNT_FLAG_DISABLE
Sent Fast Error Notification Enable	☐
Sent Fast Error Notification	NULL_PTR
Sent Slow Error Notification Enable	☐
Sent Slow Error Notification	NULL_PTR
Fast Message Read type	Interrupt
Sent Fast Notification	SentFastNotification_MLX90367LD
Slow Message Read type*	Polling
Sent Slow Notification	SentSlowNotification_MLX90367LI

▼ Sent SPC mode Support

Name SentSPCmode

Sent SPC Mode Enable	☐	Sent SPC Mode Calibration diagnosis Enable	☐
Width Of The SPC Pulse (1 -> 64)	1		
Delay Of The SPC Pulse (1 -> 64)	1		
Tick base of SPC	INT_SPC_TICK_BASE_CONFIGURED		
SPC pulse trigger type	SENT_SPC_SOFTWARE_TRIGGER		

7) Click the Generate Project to generate relevant code.

